

TAMIL HANDWRITTEN CHARACTER RECOGNITION

CS6611 – CREATIVE AND INNOVATION PROJECT

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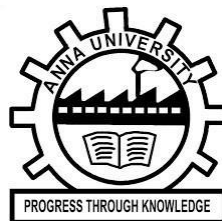
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in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



College Of Engineering, Guindy

ANNA UNIVERSITY, CHENNAI – 600 025

APRIL 2025

ANNA UNIVERSITY : CHENNAI 600 025

BONAFIDE CERTIFICATE

Certificate that this project request titled **TAMIL HANDWRITTEN CHARACTER RECOGNITION** is the bonafide work of Ashwin Arul M(2022103306), Sathish T (2022103308), Gokulakannan T (2022103708) who carried out the project work, for the fulfilment of the requirements as part of the CS6611 – Creative and Innovative Project.

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ACKNOWLEDGEMENT

Foremost, we would like to express our sincere gratitude to our project guide **Dr. C.VALLIYAMMAI**, Professor and **Dr.G.S.MAHALAKSHMI**, Associate Professor, Department of Computer Science and Engineering, College of Engineering Guindy, Chennai for their constant source of inspiration. We thank them for the continuous support and guidance which was instrumental in taking the project to successful completion.

We are grateful to **Dr. V.MARY ANITA RAJAM**, Professor and Head, Department of Computer Science and Engineering, College of Engineering Guindy, Chennai for her support and for providing necessary facilities to carry out for the project.

We would also like to thank our friends and family for their encouragement and continued support. We would also like to thank the Almighty for giving us the moral strength to accomplish our task

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ABSTRACT

Tamil Handwritten Character Recognition is a significant area of study within the broader domains of Optical Character Recognition (OCR) and Pattern Recognition. The primary objective of this research is to design accurate, robust, and efficient algorithms capable of recognizing and digitizing handwritten Tamil script.

Given the rich heritage and widespread usage of the Tamil language, creating systems that can understand handwritten input has become increasingly valuable for applications in education, digital archiving, and automation.

Recognizing handwritten characters, especially in a script as intricate as Tamil, presents a number of challenges. Variations in individual handwriting styles, inconsistencies in character shapes, and differences in writing patterns contribute to the complexity of the task. Unlike printed characters, handwritten letters often lack uniformity, and certain characters may appear similar in form, making accurate classification difficult.

Despite these challenges, recent advancements in machine learning and deep learning have significantly enhanced the performance of handwritten recognition systems. Convolutional Neural Networks (CNNs),

Recurrent Neural Networks (RNNs), and other deep learning architectures have shown great potential in extracting features, learning complex patterns, and achieving high accuracy in recognition tasks. These models can be trained on large datasets of handwritten Tamil characters, enabling them to generalize across a wide variety of writing styles.

Ongoing research continues to focus on improving model accuracy, reducing computational requirements, and expanding the scope of recognition systems to include compound characters and complete handwritten texts.

CHAPTER 1

INTRODUCTION

1.1 Introduction to Domain

Tamil Handwritten Character Recognition is a specialized area within the broader fields of Optical Character Recognition (OCR) and Pattern Recognition. It focuses on the automatic identification, classification, and interpretation of characters written in the Tamil script. As one of the world's oldest classical languages, Tamil has a rich literary history and a script known for its curvilinear and complex character structure. This uniqueness makes the recognition of handwritten Tamil particularly challenging and intellectually stimulating as a research problem.

One of the key difficulties in Tamil handwritten character recognition lies in the wide variability of handwriting. Individual writing styles differ significantly in terms of stroke thickness, character slant, spacing, and curvature. Additionally, many Tamil characters share visual similarities, increasing the risk of misclassification. These factors make it difficult to achieve consistent accuracy using traditional rule-based approaches.

The rise of machine learning and deep learning techniques has opened new possibilities for overcoming these challenges. Neural networks, particularly Convolutional Neural Networks (CNNs), have shown great promise in learning to recognize patterns and features directly from raw image data. With sufficient training data, these models can adapt to a wide range of handwriting styles, offering high recognition accuracy and robustness.

There is a growing need for handwritten Tamil recognition systems in various sectors, including education, historical manuscript digitization, banking, and authentication systems. Digitizing handwritten content can aid in preserving cultural heritage, improving accessibility to handwritten documents, and automating document processing workflows. As such, Tamil handwritten recognition is not just a technical challenge, but also a valuable tool for language preservation and digital transformation.

Overall, this domain is witnessing rapid advancements due to the integration of artificial intelligence and computational models. With continued research and development, Tamil handwriting recognition systems are expected to become more efficient, accurate, and applicable across real-world scenarios.

1.2 Motivation:

The motivation behind developing a Tamil handwritten recognition system lies in the increasing demand for automated solutions in document digitization, educational tools, authentication systems, and digital archiving. Tamil, being a language with a vast literary and cultural heritage, has a significant volume of handwritten content that remains undigitized. Moreover, many administrative and educational institutions still rely on handwritten records. By enabling machines to read and interpret Tamil handwriting, we can bridge the gap between traditional documentation and modern digital systems, thereby improving accessibility, storage, and analysis of textual data. Recent improvements in deep learning and computational power also make it feasible to tackle these challenges with greater accuracy and efficiency.

1.3 Objectives:

The main objectives of this project are as follows:

- To develop a machine learning-based algorithm that can accurately recognize and classify handwritten Tamil characters.
- To design a system that can identify and adapt to variations in handwriting styles, character shapes, and personal writing patterns.
- To ensure the algorithm is robust across different levels of handwriting quality, including poor or inconsistent strokes.
- To ultimately build a real-time Tamil handwriting recognition system that can be integrated into practical applications such as:
 - Document Analysis for digitizing handwritten manuscripts.
 - Signature Verification for banking and legal systems.
 - Handwriting-Based Authentication Systems for security and identity validation.

CHAPTER 2

EXISTING SYSTEM

Tamil handwritten character recognition has traditionally been a challenging task due to the script's complexity and the wide range of individual writing styles. In existing systems, the recognition process typically involves conventional image processing techniques and basic machine learning algorithms. While these methods have laid the foundation for further advancements, they face several limitations in terms of accuracy, scalability, and adaptability to real-world handwriting variations.

2.1 Use of Optical Character Recognition (OCR) Tools:

One of the primary approaches in earlier systems is the use of Optical Character Recognition (OCR) tools such as Tesseract OCR. These tools are highly effective when it comes to recognizing printed Tamil text; however, their performance significantly degrades when applied to handwritten characters. The core reason is that these OCR engines are designed with assumptions of uniformity and clarity, which are characteristics of printed fonts but not of human handwriting.

Moreover, traditional OCR engines lack the capability to adapt to the high degree of intra-class variability found in handwritten Tamil characters. Variations in stroke width, alignment, spacing, and curvature can cause OCR tools to misclassify or completely miss characters. As a result, their use is mostly limited to scenarios involving scanned books, documents, and newspapers — where the text is neatly printed.

2.2 Machine Learning Models with Manual Feature Extraction

Some of the early attempts to improve handwritten Tamil character recognition involved the application of traditional machine learning algorithms such as:

- **Support Vector Machines (SVM)**
- **k-Nearest Neighbors (k-NN)**
- **Decision Trees**

These models often rely on **manual feature extraction** methods, where the input images are preprocessed to extract geometric or statistical features such as:

- Histogram of Oriented Gradients (HOG)
- Zoning
- Pixel intensity histograms
- Structural features (loops, endpoints, etc.)

Furthermore, these models struggle to capture the **spatial and contextual information** that is often crucial in distinguishing similar Tamil characters. This limits their applicability in real-world scenarios where character boundaries may not be well-defined, and noise or distortions are common.

2.3 Challenges in Dataset Availability and Diversity

Another significant limitation in the existing system is the **lack of comprehensive and diverse handwritten Tamil datasets**. Many models are trained on small or limited datasets, which do not adequately represent the variability of Tamil handwriting across different age groups, regions, or writing instruments. As a result, models trained on these datasets tend to perform well in controlled environments but fail when deployed in the wild.

In addition, older systems rarely support:

- **Real-time recognition**
- **Contextual error correction**
- **Dynamic adaptability to new handwriting styles**

CHAPTER 3

System Design

3.1 System overview

The Tamil Handwritten Character Recognition system is designed to identify individual handwritten Tamil characters from scanned or digital images. The core of the system lies in using deep learning—specifically, a Convolutional Neural Network (CNN)—to learn and generalize from large datasets. The system accepts an image input, preprocesses it, feeds it into the trained model, and outputs the predicted Tamil character. The overall design ensures modularity, scalability, and ease of integration into real-world applications.

3.2 Flow Diagram

To illustrate the functional structure of the Tamil Handwritten Character Recognition System, Figure 3.2 presents a comprehensive architecture diagram. It outlines the key modules and their interactions across various functional layers, offering a clear perspective to both developers and evaluators. The diagram begins with the **Login** module, branching into core functional areas such as **Practice Letters**, **Practice Words**, **Tamil Numbers**, and **Parents Tracking**. Each path includes dynamic interactions — from difficulty levels to prediction confidence rates — culminating in character analysis and user progress monitoring. Arrows represent data flow and decision-making logic, guiding users through personalized learning, while the **Parents Tracking** module aggregates performance metrics across letters, words, and numbers. This architectural visualization serves as the blueprint of the system, demonstrating how handwriting input leads to prediction, evaluation, and feedback in an intuitive and adaptive environment.

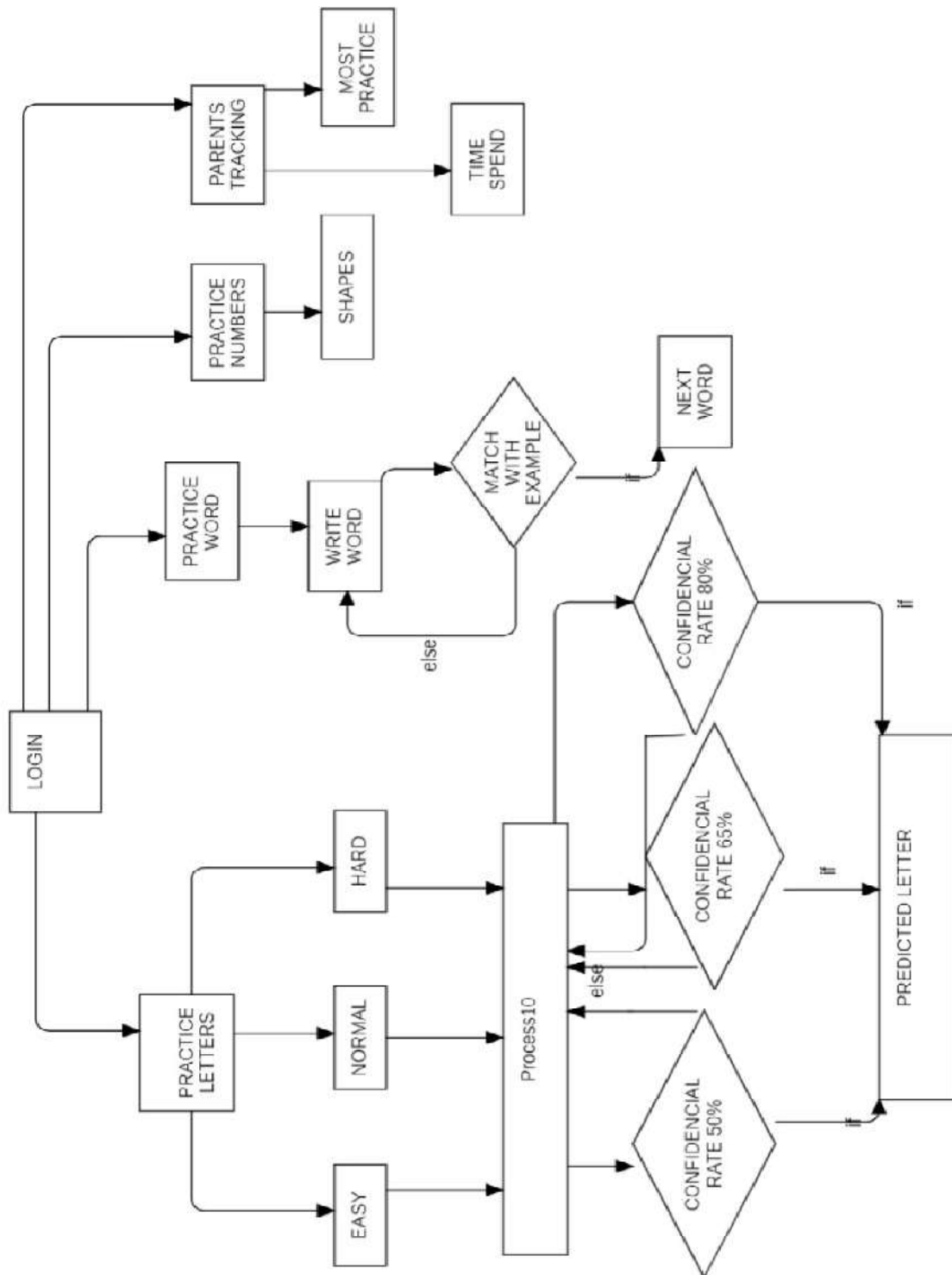


Figure 3.2 (Flow diagram)

3.2.1 Overview of the System Flow

The architecture diagram is the foundation of the Tamil Handwritten Character Recognition System, providing a visual illustration of the flow of control, module interaction, and the structure that facilitates user interaction and learning. The system is well-planned to meet the needs of early-stage learners, particularly children or non-native speakers, through the provision of a structured and interactive environment for the practice of Tamil letters, words, and numbers. Addition of intelligent feedback systems and performance monitoring makes the platform not only a learning utility but also turn into an adaptive tutor for the user.

At the entry point of the system is the Login module, which initializes the user session and loads per-student progress data. From here, the user can proceed to various modules like Practice Letters, Practice Words, or Tamil Numbers. Each practice route leads to a writing interface where the student types Tamil characters with a stylus or mouse. The system takes this input and processes it using handwriting recognition models to recognize the character and evaluate the degree of similarity to the expected output.

For character recognition, the system applies a layered confidence-based prediction model, which is one of the primary characteristics of the system's intelligence. It assesses the handwritten input across different confidence levels (e.g., 50%, 65%, 80%), giving users immediate feedback and leading them towards improved accuracy. If the prediction aligns with the anticipated character with adequate confidence, it is identified as correct; otherwise, the user is asked to retry. This loop maintains proper writing habits and establishes muscle memory through repetition.

Also incorporated is a Parents Tracking module, which gives information to parents and teachers about the learner in depth. This comprises summaries of performance on letters, words, numbers, and practice time in total. The design incorporates a persistent learning cycle where input, prediction, feedback, and tracking become an integrated process directed toward mastery of Tamil handwriting.

3.2.2 Login Module

The Login Module is the first point of entry into the Tamil Handwritten Character Recognition System. This key module authenticates and identifies users prior to

accessing customized features in the application. The login module is made intuitive and secure to provide a smooth user experience while protecting sensitive information.

User Authentication

When the application is launched, users are asked to provide their credentials (password and username). These credentials are checked against a secure database. The system uses industry-standard encryption methods to make sure that user data is kept confidential while it is being transmitted and stored. If the user is registered, the system checks their identity by comparing the input credentials with the stored information. If the credentials are invalid, users are given a prompt to re-enter the details.

Customization and Personalization

After the user logs in successfully, the system adjusts the experience for individual learning requirements. The Login Module is an entry point to customized features like:

Practice Letters: The user is free to start practicing Tamil characters, with difficulty levels set as Easy, Normal, and Hard.

Practice Words: Users can practice words that consist of the letters they have learned to master.

Practice Numbers: A section where Tamil numerals are practiced.

This customized approach ensures that the learning pathway of each user is tailored according to his or her development and interests.

Data Monitoring and Tracking

The Login Module also provides for monitoring user performance across time. All interactions with the system are tracked, including:

Duration spent on each practice session (letters, words, or numbers).

Recognition and writing accuracy rates of Tamil characters.

Use frequency and progress on different difficulty levels.

These statistics are recorded in a database and associated with the user's account, enabling them to view their history of performance. This aspect promotes ongoing learning through providing feedback on areas that need improvement.

Session Management

For improving user experience, the Login Module has session management support, enabling users to save their progress and resume from where they left off in future sessions. Users can also log out at any time, which securely ends their session and safeguards their data.

User Registration and Profile Creation

For the new users, the Login Module provides a feature to register for a new account. While registering, users are prompted to enter necessary details like their name, email address, and a strong password. Once registered, an email is sent to the user to confirm the account. Once confirmed, the user can log in with their created credentials.

Security Features

Security is of primary concern to the Login Module. Beyond the regular username and password authentication, the system can additionally have multi-factor authentication (MFA) available for extra safeguarding, so users must verify through an alternative means (for example, using a mobile app or email). This guarantees authorized users alone get access to the system.

3.2.3 Practice Letters Module

The Practice Letters Module has been created to enable users to enhance their mastery in writing Tamil characters by providing them with exercises suited to their skill level. This module has been designed as a progressive learning process, organized into three difficulty levels: Easy, Normal, and Hard. Each level has been designed to challenge users in distinct ways to encourage gradual progress and mastery over handwriting Tamil.

3.2.4 Level Breakdown

3.2.4.1 Easy Level:

The Easy Level deals with the most common Tamil letters, which are easy to write and identify. They are generally the root characters and comprise basic vowels, consonants, and simple blends. The Easy Level is meant for beginners who are new to learning the Tamil script. The main purpose at this level is to introduce users to the shapes and strokes of letters.

Example Characters:

- □ (a)
- □ (e)
- □ (ka)
- □ (sa)

3.2.4.2 Normal Level:

The Normal Level provides more advanced Tamil letters that need better comprehension of the intricacies of the script. The level employs medium complexity characters that include extra curves, strokes, and linkage between the letters. It is ideal for those who have understood the fundamentals and are now set to move a step ahead in their skills.

Example Characters:

- த (tha)
- ம (ma)
- ன (na)
- ல (la)

3.2.4.3 Hard Level:

The Hard Level contains the most difficult characters of the Tamil script for advanced students who have completed the basic and intermediate levels. These characters generally have complex combinations of strokes and more refined variations in shape.

Example Characters:

- ற (ra)
- ழ (zha)
- ஞ (ksha)
- ஜ (ja)

Interactive Learning Environment

Every level in the Practice Letters Module sends the user to the Write Tamil Characters interface, where they are shown a character to copy. This interface is created to provide an interactive learning environment, with:

Character Display: The program displays the character that the user must write on the screen. Users can also view a clean, step-by-step guide on how to write the character if necessary.

Writing Space: A special writing space where users can trace or write the Tamil character freely using a stylus (on a touch screen) or a mouse. This function replicates actual handwriting, providing a more natural learning experience.

Feedback Mechanism: Upon typing out the character, the system applies handwriting recognition technology to scan the input. When the character typed out is in the correct form, the user is given positive feedback (e.g., "Well Done!"). When the character is typed out in an incorrect manner, the system gives constructive feedback, such as identifying certain errors in strokes or shapes.

Real-Time Guidance

To help in writing characters, the module offers real-time feedback in the form of visual aids, including:

Tracing Mode: At the initial stages, the system provides a tracing mode where the user can trace the provided character, enabling them to become familiar with the strokes.

Stroke Order Guidance: For certain of the more complicated characters, the system may offer an outline or stroke order animation, assisting users in comprehending the order in which the strokes are to be drawn.

Confidence Level: After the user posts their written character, the system assesses its correctness and issues a confidence level representing how well the written character is done in comparison to the correct form. This helps motivate the user to practice repeatedly until they reach high accuracy.

Progress Tracking and Encouragement

As the user advances through the levels, performance is monitored continuously. The system keeps track of:

Accuracy Rate: The rate at which characters are written correctly.

Writing Speed: How long it takes to write every character or combination of characters.

Consistency: Whether the user is able to reproduce the character repeatedly over time.

This information is saved and can be retrieved through the user's profile for future use. To encourage repeated practice, the system provides badges or rewards upon reaching certain milestones, like getting through a series of characters in a specific level with high accuracy or speed.

3.2.5 Character Recognition & Prediction

The Character Recognition & Prediction module is the focus of the Write Tamil Characters interface as it takes the user's handwritten input and gives real-time feedback on the accuracy of its recognition.

Handwritten Input Capture

After the user has written a Tamil character in the provided writing area, the system captures the input through a stylus or mouse. The input is transformed into a digital form that the system can analyze. This representation is processed via a sequence of image pre-processing operations like noise reduction, resizing, and normalization, to increase the accuracy of the recognition process. The system guarantees the captured image to be clear, readable, and oriented for optimal analysis.

Pattern Recognition

The foundation of the recognition process hinges on pattern recognition algorithms that compare the captured handwriting character with an enormous database of pre-defined patterns of Tamil characters. The pattern recognition algorithms implement machine learning models that have been trained:

Feature Extraction: The system recognizes primary features of the character, i.e., curves, lines, and intersections, which are exclusive to every Tamil letter.

Matching Algorithm: Extracted features are compared with the nearest available character template in the system database. This is based on distance metrics, e.g., Euclidean distance or cosine similarity, for the best fit.

Confidence Evaluation

After the system processes the input and matches it against the stored templates, it returns a confidence score that is an indication of how closely the input resembles a specific character. The confidence score is a percentage between 0% and 100%, representing the degree of certainty of the system's prediction. The confidence assessment is conducted in several stages:

Confidence Rate 50%

At this early phase, if the system realizes that the match has a 50% confidence level, then it means the recognition is provisional. At this level, the system marks the result as dubious, and the user is requested to try the character again. The user is motivated to practice handwriting to raise the system's confidence in reading the character correctly.

Confidence Rate 65%

If the confidence in recognition is at 65%, the system will rate the match as moderately trustworthy. Although the outcome is partially correct, the system can still be uncertain of the final guess. In this instance, the user might be prompted to inspect the guess and either approve or enter a new input. The system can provide hints or graphical guides to help the user correct their handwriting.

Confidence Rate 80%

Under 80% confidence level, the system's identification is extremely more precise, while the prediction for the character would be highly regarded as trustworthy. When the predicted value surpasses the 80% threshold value, the system identifies the character as Correct with the revealed anticipated letter as seen for validation purposes. The established confidence level confirms that the system is near in its surety of recognition.

3.2.6 Practice Words Module

In this module, users are prompted to **Write Tamil Characters** that make up words. The system compares each word with a stored **example word**. If the match succeeds, the flow proceeds to **Next Word**, creating a continuous learning experience. In case of mismatch, it offers retries or hints

3.2.7 Tamil Numbers Module

The **Tamil Numbers** module focuses on helping users learn and **Count Numbers** in Tamil script. This allows for number writing practice in a similar recognition-based process, enabling learners to master numeric forms alongside alphabetic ones.

3.2.8 Parents Tracking Module

A key part of the system is the Parents Tracking Module, which provides real-time information about a student's performance and learning activity. The module equips guardians with the ability to monitor different facets of the student's learning process, keeping them informed about their child's performance.

Words: Besides isolated letters, the module also keeps tabs on how the student performs while writing and reading words. These are how many words they've practiced, correctness in spelling, and total writing fluency in joined-up letters.

Numbers: For Tamil numeral practice, the module logs how well a child performs reading and writing numbers and enables parents to determine how good the child is at dealing with various sets of characters.

Spending Time: Parents have the opportunity to see how many hours their child spends studying letters, words, and numbers. This tool assists in assessing the productivity of the child's study time and makes sure that the learning time is productive and balanced.

3.2.9 Confidence Evaluation Logic

Confidence Evaluation Logic is a core component of the system that verifies every character typed by the user and assesses it through various confidence levels. The trained model of the system employs sophisticated pattern recognition methods to analyze the user's handwriting input against standard character templates. The system determines a confidence level based on how well the user's handwriting input matches the character templates.

The confidence score plays a significant role in deciding whether the character is properly written. The system works with various confidence levels, normally from low to high, which determine the acceptance or rejection of the written character. At every step, the system compares the input with a given confidence level. If the system attains a minimum confidence level—usually 50%, 65%, or 80%—the character is deemed to be a match, and either the system accepts the input or asks the user if they want to confirm.

Whenever the confidence score goes below the lower limit, the system initiates a recursive attempt, asking the user to resubmit the character. Such an iterative feedback system is planned to motivate refinement. By having the user edit their writing, the system achieves the goal that learning is improved progressively. Each repeat attempt presents the user with the chance of rectifying faults and improving awareness about the development of the Tamil characters.

3.2.10 Feedback and Looping Flow

The Feedback and Looping Flow is an essential component of the adaptive learning process of the system. Every primary activity, be it writing words, letters, or numbers, involves a well-structured feedback loop based on the recognition outcomes. After the user writes a character or a word, the system processes the input and matches it with pre-defined templates through pattern recognition algorithms. When the match is successful, the system gives positive feedback to verify that the input is accurate and the user can go to the next task.

However, in case the match is unsuccessful or does not attain the specified confidence level, the system transitions into a correction stage. Here, the user will be asked to rewrite the word or character again. The system can offer prompts or visual cues to assist the user to correct their handwriting. This guarantees that errors are corrected at once, and learning is reinforced through practice.

3.2.11 Integration and User Experience

The system integrates different elements in a seamless manner to provide an integrated and seamless user experience. Users interact with a drawing interface on the frontend, where they can enter handwritten characters, and the predictions are displayed in real-time. The backend is responsible for processing handwriting by employing pattern recognition algorithms to interpret the input from the user and perform predictions calculations. Backend logic plays a significant role in ensuring that the characters are recognized correctly by the system and adequate feedback is generated.

3.3 Algorithm Workflow

1. User Input

- The user writes a Tamil character (letter, word, or number) on the drawing canvas.

2. Capture Input

- The system captures the handwritten input as a digital image.

3. Preprocessing

- Image preprocessing is applied: noise reduction, resizing, and normalization.

4. Pattern Recognition (Backend)

- The system applies machine learning models to analyze the preprocessed input and extract features.

5. Confidence Evaluation

- The system computes the confidence score of the predicted character.
- Decision:
 - If Confidence Score $\geq$ Threshold (e.g., 80%), Proceed to Step 7 (Feedback to User).
 - If Confidence Score $<$ Threshold, Proceed to Step 6 (Retry Input).

6. Retry Input (if Confidence $<$ Threshold)

- Prompt the user to reattempt the character.
- Provide visual guidance or hints.

7. Feedback to User

- If Prediction is Correct:
 - Display predicted character and confirm success.
 - Proceed to next task or learning module.
- If Prediction is Incorrect:
 - Display error message and prompt for reattempt.

8. Progress Tracking (Database)

- Store the user's progress: accuracy, time spent, and learning session data.

9. End of Session

- After completing a set (e.g., letters, words, numbers), update progress in the database.

3.4 Architecture diagram

This diagram outlines the process of recognizing handwritten characters using machine learning. It starts with **data acquisition** and **input of handwritten characters**, followed by **data preprocessing** to clean and prepare the input. Then, **features are extracted and selected**, and a suitable model is chosen and trained. The model makes **predictions**, and the result is displayed through the **user interface**. The process ends after output is delivered.

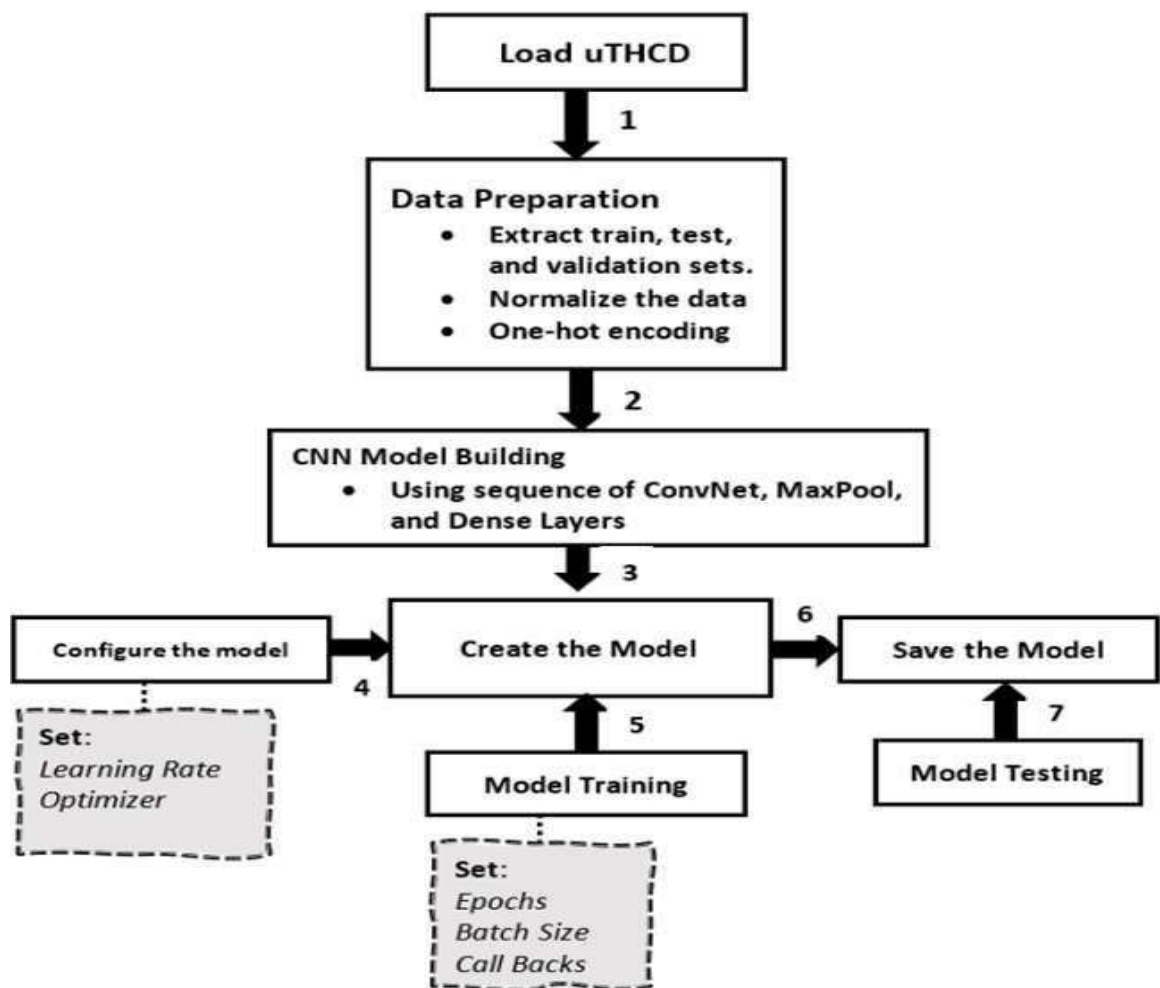


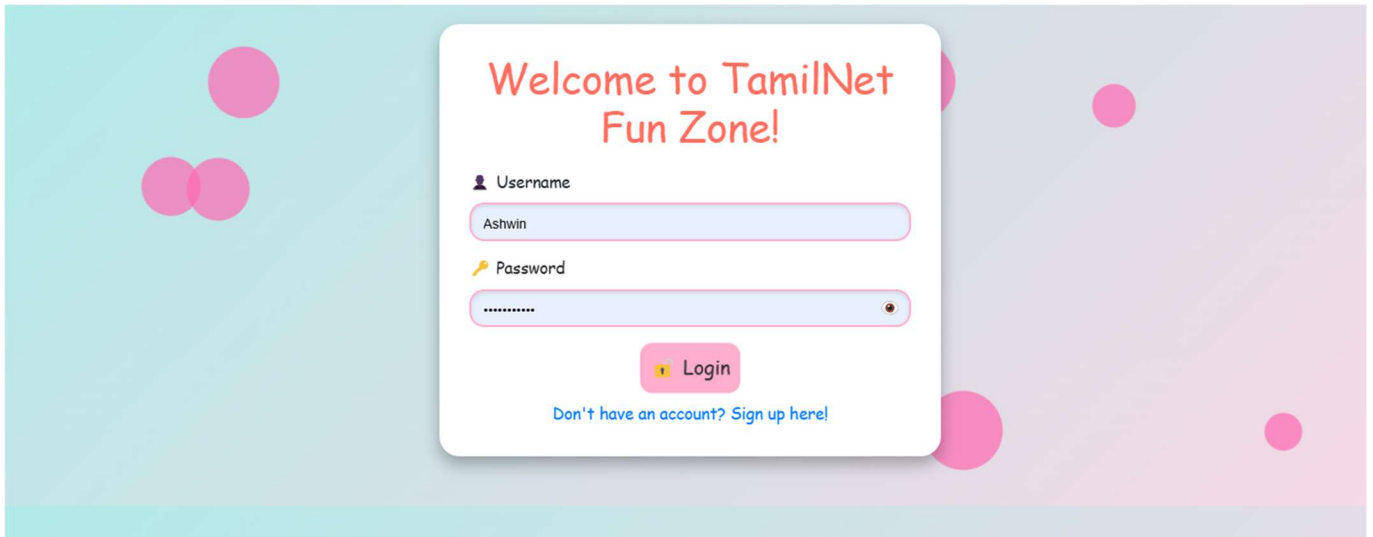
Figure 3.4

CHAPTER 4

IMPLIMENTATION

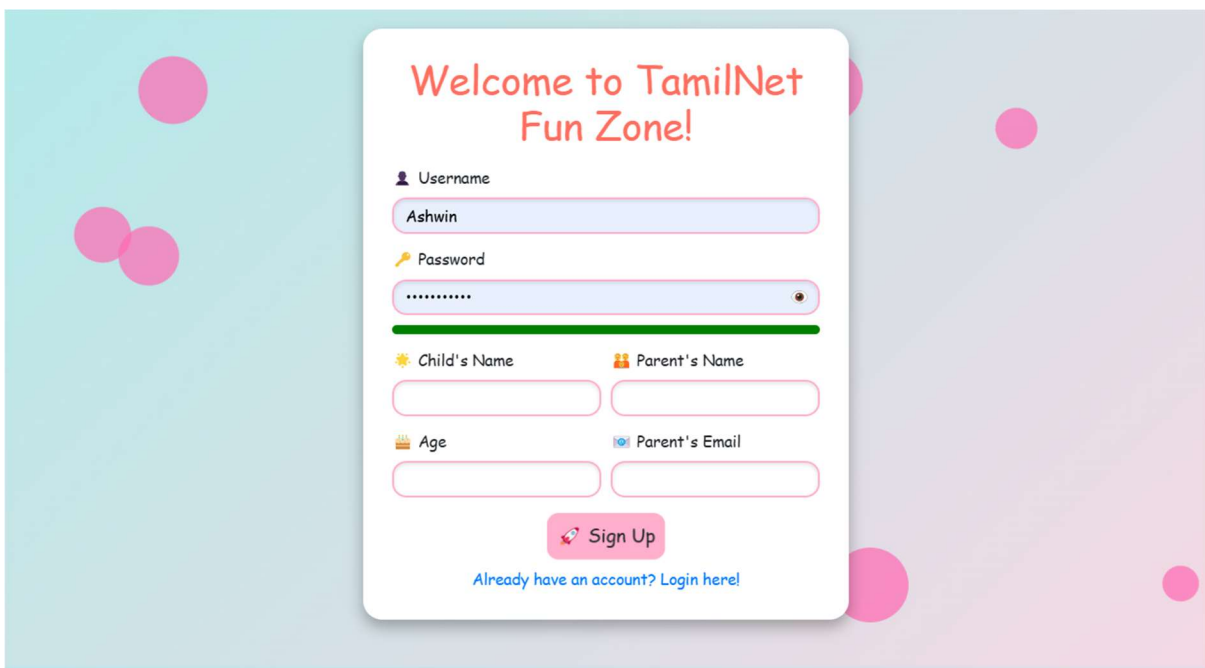
4.1 : Login page

Figure 4.1



The login page features a white central card on a light blue and pink background with large pink circles. The card has the title "Welcome to TamilNet Fun Zone!" in red. Below the title are two input fields: "Username" with the value "Ashwin" and "Password" with masked characters "*****". A pink "Login" button is positioned below the password field. At the bottom of the card, there is a link: "Don't have an account? Sign up here!"

4.1.2: Sign up page



The sign up page features a white central card on a light blue and pink background with large pink circles. The card has the title "Welcome to TamilNet Fun Zone!" in red. Below the title are two input fields: "Username" with the value "Ashwin" and "Password" with masked characters "*****". A green horizontal line separates the login fields from the registration fields. The registration section includes four input fields: "Child's Name", "Parent's Name", "Age", and "Parent's Email". A pink "Sign Up" button is positioned below the registration fields. At the bottom of the card, there is a link: "Already have an account? Login here!"

Figure 4.1.2

4.2 Main Dashboard – Tamil Character Fun Zone

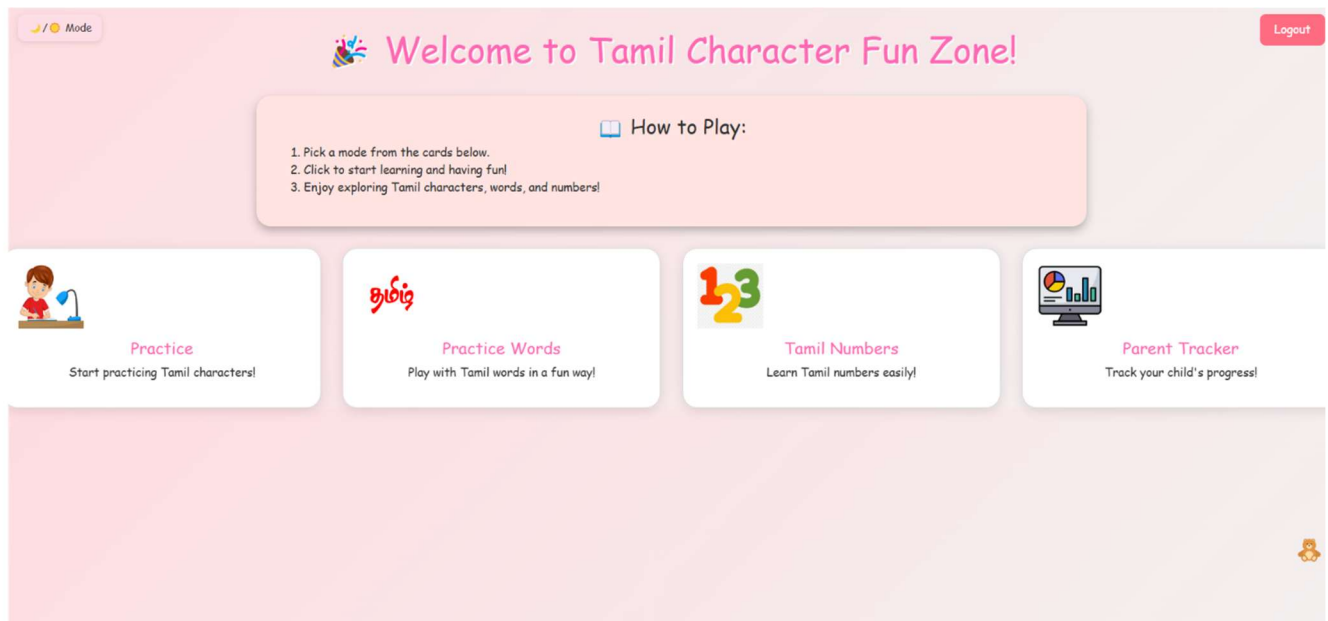


Figure 4.2

The Main Dashboard is the homepage that appears after a successful login. It welcomes users with the title: **"Welcome To Tamil Character Fun Zone!"**

There are 4 colorful cards for users to choose from:

- Practice – Practice individual Tamil letters.
- Practice Words – Learn and practice simple Tamil words.
- Tamil Numbers – Learn Tamil numbers in a fun way.
- Parent Tracker – Parents can check their child's progress.

There's also a Logout button at the top right to exit safely.

The whole page is made bright and friendly to make learning fun and easy for kids

4.3 Practice Area

The Tamil Character Fun Zone is a playful and educational tool designed to help kids learn to draw Tamil characters through a gamified approach.

Game Modes:

The system features three difficulty levels:

- Easy
- Medium
- Hard

Each mode adjusts the character recognition model's expected accuracy, encouraging learners to progress gradually.

4.3.1 Practice Area – Easy mode



Figure 4.3.1

In this mode, the model operates at a basic recognition level with an expected accuracy of 50%. This is suitable for beginners and young learners who are just starting out and need a forgiving environment to explore Tamil writing

4.3.2 Practice Area – Medium mode



Figure 4.3.2

The model’s prediction improves in this mode with an expected accuracy of 65%. It is ideal for learners with some familiarity, encouraging more accurate drawing while still allowing for minor mistakes

4.3.3 Practice Area – Hard mode



Figure 4.3.3

This mode challenges users with a more strict prediction threshold. It uses an expected accuracy of 80%, suitable for more experienced users aiming to perfect their Tamil character writing.

4.4 Practice Word

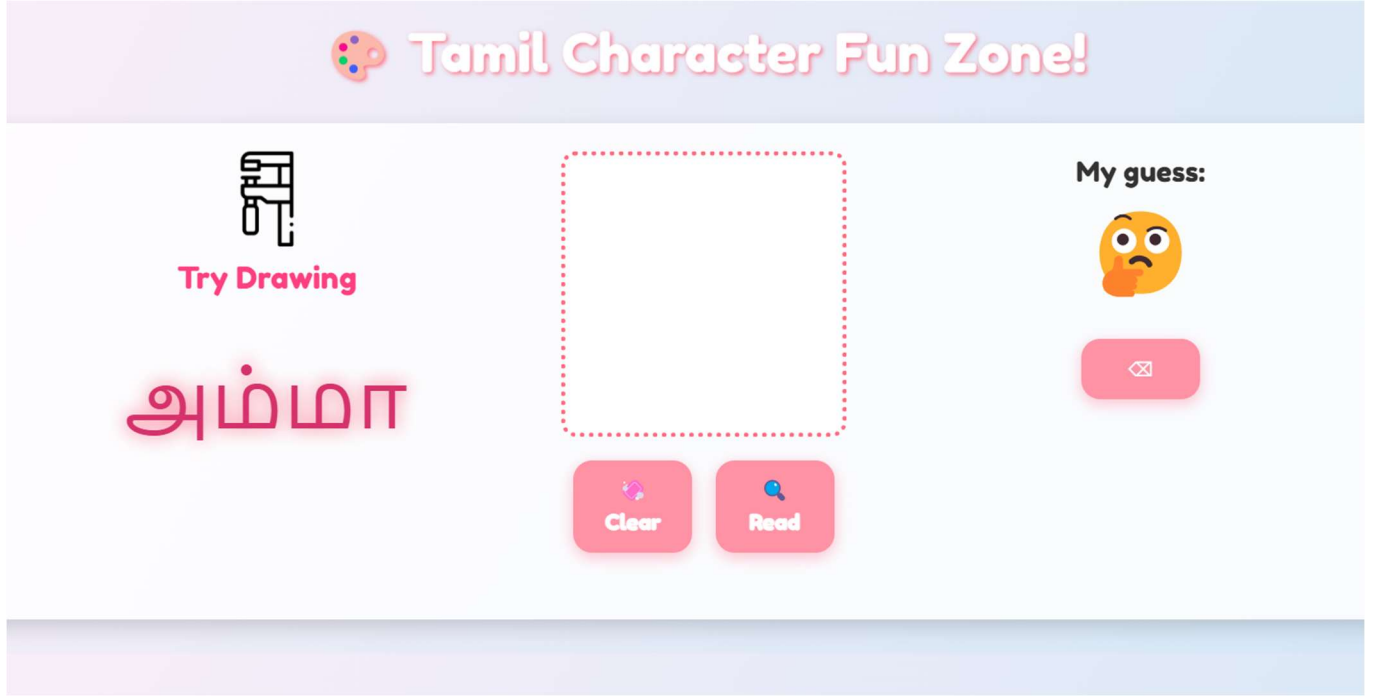


Figure 4.4

This screen from TamilNet for Kids helps children learn Tamil by practicing how to write words. The word “அ ம்மா” and various words are shown, and kids try to draw it in the center box. The app then tries to guess the word using handwriting recognition. If it can’t guess, a thinking emoji appears. The colorful design and simple layout make it fun and easy for kids to use.

4.5 Practice Tamil Numbers – Home

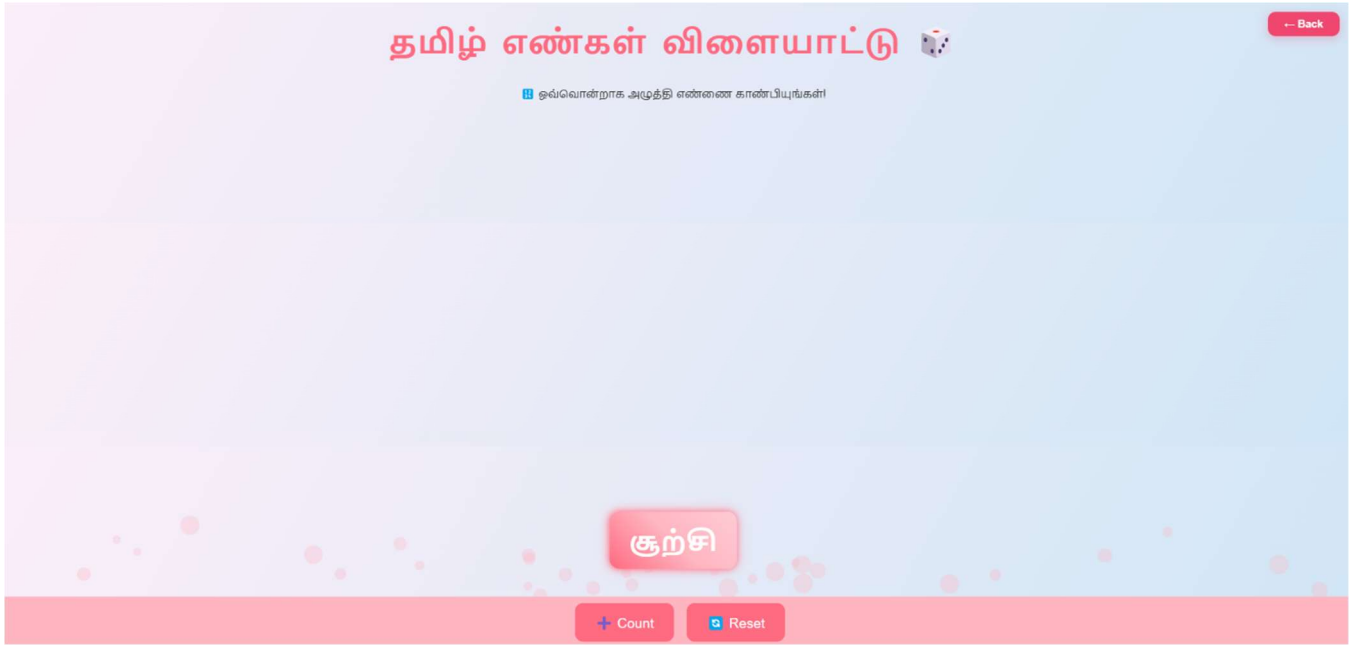


Figure 4.5.1

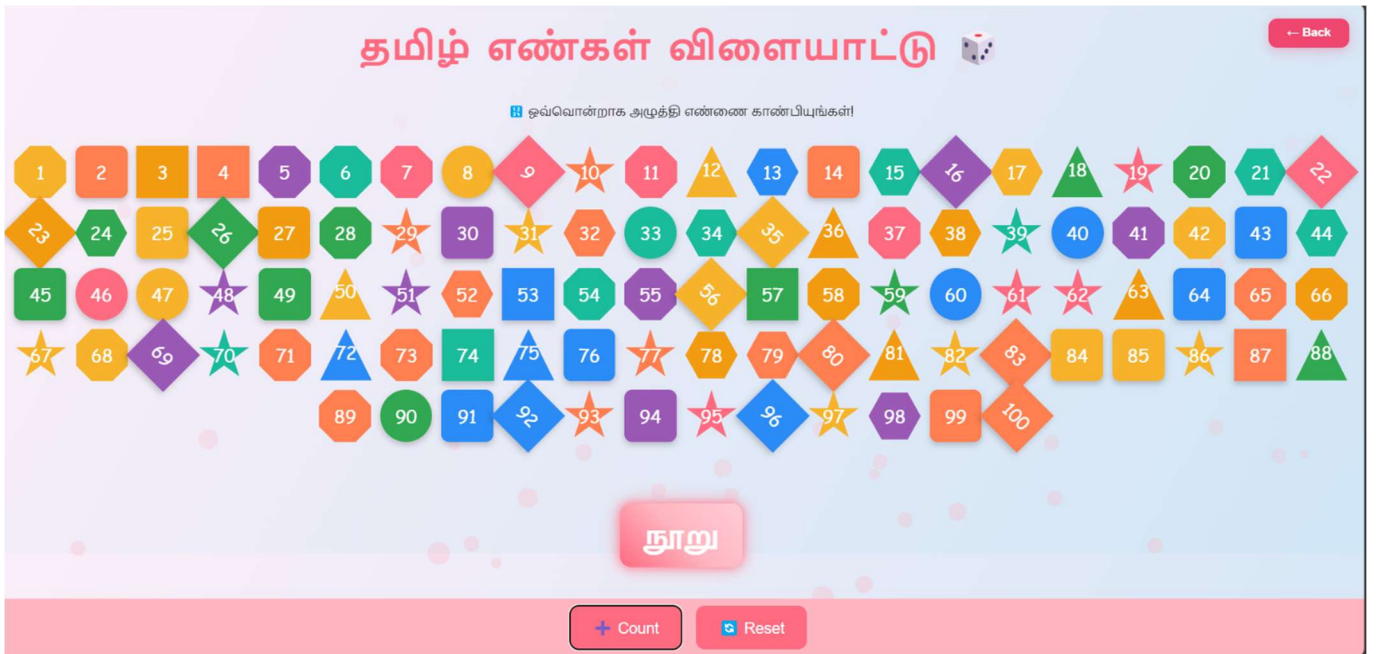


Figure 4.5.2

Kids can click the “Count” button to increase the number shown in Tamil. Each time they click, the number updates — for example, " ௦ " (zero), " ௧ " (one), and

so on. There's also a “Reset” button to start over. The design is simple and colorful, making learning easy and enjoyable for young learners.

4.6 Parent Dashboard

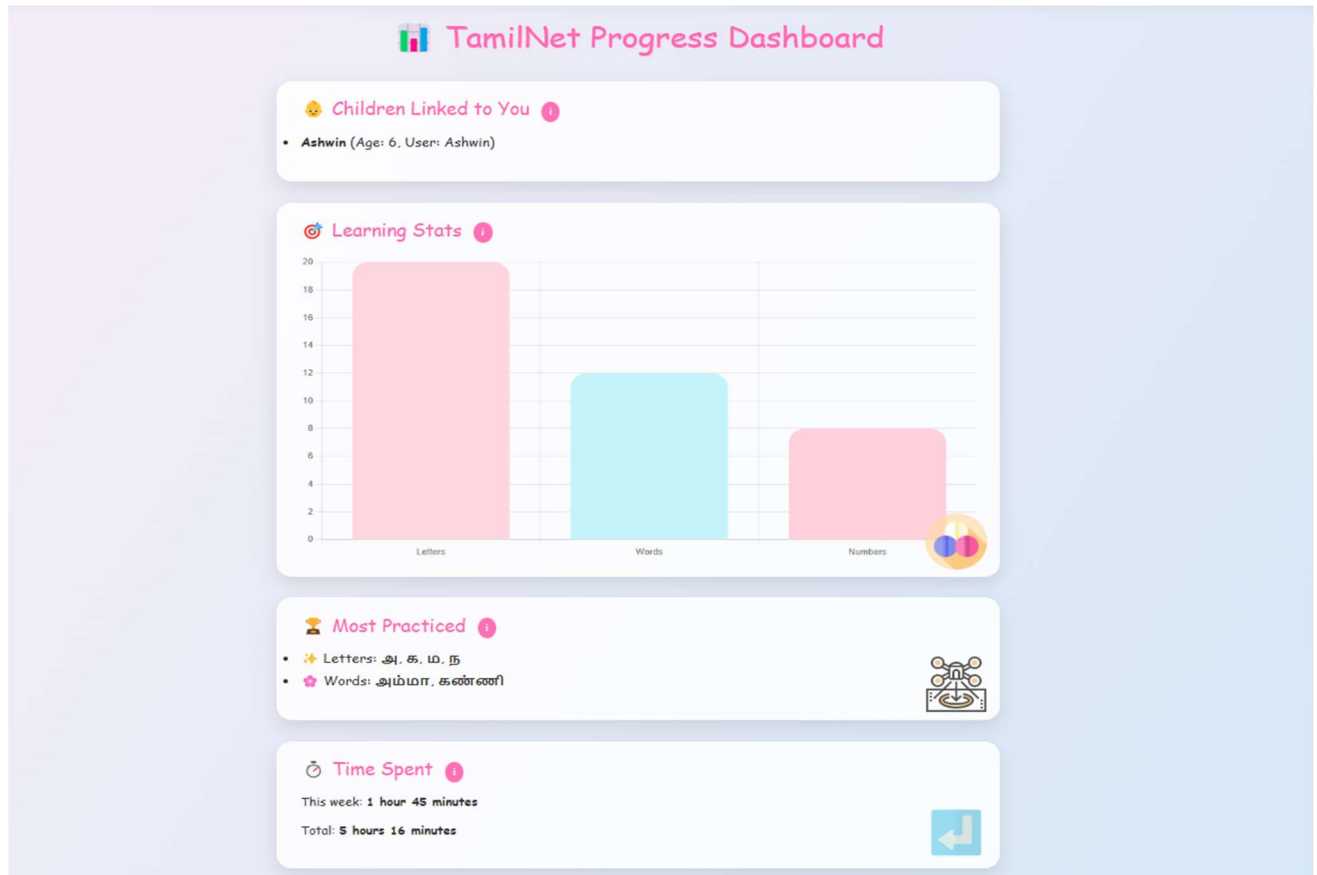


Figure 4.6

The dashboard designed for parents to track their child’s learning progress in Tamil. It shows a colorful bar chart with stats on how many letters, words, and numbers the child has practiced. Below the chart, it displays the most practiced letters and words. At the bottom, it also shows the time spent learning during the week and the total time so far. The soft colors and clean layout make it easy for parents to understand their child’s progress at a glance.

CHAPTER 5

PERFORMANCE METRICS

Model Performance Report

Project Title: Tamil Handwritten Character Recognition using CNN

Dataset Used: TamilNet (Train / Validation / Test Splits)

Number of Classes: 156 Tamil Characters

Model Architecture: Convolutional Neural Network (CNN)

1. Training Metrics

Metric	Value Range	Final Value (Epoch 30)
Training Loss	Decreased from 1.9 to 0.55	0.55
Training Accuracy	Increased from 35% to 90.5%	90.5%

2. Validation Metrics

Metric	Value Range	Final Value (Epoch 30)
Validation Loss	Decreased from 2.0 to 0.77	0.77
Validation Accuracy	Increased from 30% to 78.5%	78.5%

3. Test Evaluation

Metric	Value
Test Accuracy	90.7%
Test Dataset	TamilNet Test Set

4. Model Architecture Overview

Input: 64x64 grayscale character images (preprocessed)

Convolutional Layers:

3 convolutional blocks, each followed by Batch Normalization, ReLU activation, and Max Pooling

Fully Connected Layers:

Dense layers with Batch Normalization and ReLU activation

Final output layer with 156 output units (one for each Tamil character)

Weight Initialization: Kaiming Normal

Optimizer: Adam (Learning Rate: 0.001, Weight Decay: 0.003)

Loss Function: Cross Entropy Loss

5. Performance Summary

The model shows steady convergence during training.

Validation metrics follow a similar trend, suggesting minimal overfitting.

High test accuracy (90.7%) confirms the model's effectiveness and generalization capability.

Suitable for integration into Tamil character recognition tools or educational applications.

CHAPTER 6

CONCLUSION

Tamil Handwritten Character Recognition is a significant area of research with widespread practical applications across domains such as document digitization, signature verification, form processing, and text-based authentication systems. As Tamil is a classical language with a rich script structure and intricate characters, developing a system capable of accurately recognizing handwritten Tamil remains both a technically demanding and culturally valuable task.

Over the years, researchers have employed a variety of techniques to tackle this challenge. Traditional machine learning methods like Support Vector Machines (SVM) and Multi-Layer Perceptrons (MLP) laid the groundwork, while more recent advancements in deep learning have enabled the use of powerful architectures such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Convolutional Recurrent Neural Networks (CRNNs). These models can automatically extract and learn features from data, allowing for improved accuracy and better generalization across diverse handwriting styles.

Despite this progress, Tamil handwritten recognition continues to face challenges due to the complexity of the script, the similarity between certain characters, and the diversity of handwriting patterns among individuals. However, with the increasing availability of large and annotated datasets, as well as advancements in computing power and model optimization techniques, the development of highly accurate and efficient .

CHAPTER 7

REFERENCES

1. Shaffi, N., & Hajamohideen, F. (2021). *uTHCD: a new benchmarking for Tamil handwritten OCR*. *IEEE Access*, 9, 101469-101493.
2. I. Mahadevan, *Early Tamil Epigraphy From the Earliest Times to the Sixth Century AD (H. O. Series)*, vol. 62. Cambridge, U.K.: Harvard Univ. Press, 2003.
3. S. B. Steever, "Tamil writing," in *The World's Writing Systems*. New York, NY, USA: Oxford Univ. Press, 1996, pp. 426–430.
4. L. Campbell and V. Grondona, "Review: *Ethnologue: Languages of the world* by Raymond G. Gordon," *Language*, vol. 84, no. 3, pp. 636–641, 2008
5. R. Tamilselvi and A. Shanavas, "A survey on offline handwritten character recognition for Tamil script," *International Journal of Computer Applications*, vol. 42, no. 1, pp. 30–35, March 2012.
6. C. V. Jawahar, M. N. S. S. K. Pavan Kumar, and P. Bhat, "OCR for printed Tamil text using deep learning," *Proceedings of the 12th IAPR International Workshop on Document Analysis Systems (DAS)*, pp. 106–111, 2016.
7. S. Arulkumar and K. Vani, "Tamil Handwritten Character Recognition using Convolutional Neural Networks," *Materials Today: Proceedings*, vol. 37, pp. 3195–3200, 2021.
8. T. R. Rajeswari and M. R. Sumathi, "Offline Tamil Handwritten Character Recognition Using Deep Belief Networks," *Wireless Personal Communications*, vol. 111, pp. 1487–1503, 2020.